MIL-S-36438C
12 November 1976
SUPERSEDING
MIL-S-36438B
2 May 1975

MILITARY SPECIFICATION

SUCTION AND PRESSURE APPARATUS, SURGICAL,
Explosion-Proof, Portable, Field

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a light-weight, explosion-proof, suction and pressure apparatus mounted in a rugged, field type carrying case.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

FEDERAL

| | |
|---|---|
| QQ-B-613 | - Brass, Leaded and Non-Leaded (Plate, Bar, Sheet and Strip). |
| QQ-B-626 | - Brass, Leaded and Non-Leaded, Rod, Shaped, Forgings, and Flat Products with Finished Edges (Bar and Strip). |
| TT-E-529 | - Enamel, Alkyd, Semi-Gloss. |
| TT-P-1757 | - Primer, Coating, Zinc Chromate, Low-Moisture Sensitivity. |
| PPP-B-601 | - Boxes, Wood, Cleated-Plywood. |
| PPP-B-636 | - Boxes, Shipping, Fiberboard. |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, Code ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6515

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|---|---|
| MIL-C-5541 | - Chemical Conversion Coatings on Aluminum and Aluminum alloys. |
| MIL-M-7298 | - Manual, Technical, Commercial Equipment. |
| MIL-L-10547 | - Liners, Case and Sheet Overwrap, Water Vaporproof or Waterproof, Flexible. |

STANDARDS

FEDERAL

| | |
|---|---|
| FED-TEST METHOD STD-151 | - Metals, Test Methods. |
| FED-STD-595 | - Colors. |
| FED-TEST METHOD STD-601 | - Rubber, Sampling and Testing. |

MILITARY

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | - Marking for Shipment and Storage. |
| MIL-STD-130 | - Identification Marking of U.S. Military Property. |

DRAWING

Defense Personnel Support Center

| | |
|---|---|
| 24004 | - Suction and Pressure Apparatus, Surgical, Field. |

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

* American National Standards Institute (ANSI)

| | |
|---|---|
| ANSI C73.11 | Dimensions of Attachment Plugs and Receptacles - General Purpose 125 Volts, 15 Amperes, 2 pole, 3 wire Grounding Type. |

(Application for copies shall be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)



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American Society for Testing and Materials (ASTM)

B 135 Specification for Seamless Brass Tube

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Fire Protection Association (NFPA)

NFPA No. 56A Standard for Use of Inhalation Anesthetics

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02210.)

Underwriters' Laboratories, Inc. (UL)

U.L. 674B Electric Motors and Generators, For Use in Hazardous Locations, Class I, Groups C and D

U.L. 1067 Electrically Conductive Equipment and Materials for Use in Flammable Anesthetizing Locations

(Application for copies should be addressed to Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611; 1655 Scott Boulevard, Santa Clara, CA 95050; 333 Pfingsten Rd., Northbrook, IL 60062; 1285 Walt Whitman Rd., Melville, L.I., NY 11746).

3. REQUIREMENTS

3.1 First article. When specified in the procurement document, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.1.2.1 and 6.2).

* 3.2 Fire and casualty hazards.

3.2.1 The contractor shall submit proof, prior to delivery, that the items to be delivered under the contract conform to the applicable requirements of Underwriters' Laboratories, Inc. (U.L.) Standards of Safety 674B and 1067. Proof shall consist of one of the following:

A. U.L. label affixed to the product, provided the contractor agrees to have Underwriters' Laboratories, Inc. forward a copy of the U.L. report directly to the Contracting Officer, upon the latter's request. The unit to be submitted for U.L. testing shall be modified as follows:

1) Glass bottles shall be used.
2) No plug shall be attached to the electric cord.



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B. Laboratory report from a nationally recognized independent laboratory. If an independent laboratory is used, it must be adequately equipped and competent to determine compliance with U.L. standards. The Government reserves the right to require a facilities report and to conduct a facilities survey. Any laboratory report submitted shall include actual test methods, data taken, and results of tests. Detailed findings of examinations necessary to determine compliance shall also be included. The test report shall also have a section that describes the test sample in such a way that the report can be used to insure identity of production items with the test sample. This section shall be supplemented with construction information such as materials used, spacings between electrical components, types of bushings, photographs, wiring diagrams, sketches, and parts list, all appropriately identified. Compliance with this paragraph does not excuse compliance with all other requirements of the contract and specifications.

3.3 Material. All materials shall be in accordance with DPSC drawing No. 24004, Bill of Material.

3.4 Construction. Construction shall be in accordance with DPSC drawing No. 24004. The suction and pressure apparatus shall conform to 3.2, and shall operate on 115 volts, 60 hertz alternating current. It shall consist of a motor-driven, rotary compressor, a suction system operating through bottles in series or singly, an air filtering system providing an outlet for spray pressure, a means of protecting the pump from overflow of drained fluids, and a noise suppressor. All components shall be properly connected and provided with the proper accessories, and shall be compactly mounted within an aluminum case with a hinged aluminum cover attached. Two drainage bottles shall be securely mounted within the hinged cover in a manner permitting easy removal for cleaning purposes. Each bottle shall be fitted with an airtight rubber closure having metal couplings to fit the corresponding couplings on the rubber tubing. The apparatus shall be electrically conductive when tested as specified in 4.4.3.1 and shall pass the dielectric strength test specified in 4.4.3.4.

3.4.1 Shock and vibration. The unit in its case shall withstand, without damage, the shock and vibration test specified in 4.4.3.2. After shock and vibration test, the unit shall meet the performance requirements specified in 3.4.11.

3.4.2 Water resistance. The apparatus shall withstand the water resistance test specified in 4.4.3.3. Any entry of water shall be cause for rejection.

3.4.3 Humidity. When tested as specified in 4.4.3.6, the unit shall withstand the conditions of high humidity and temperature without damage and shall meet the performance requirements specified in 3.4.11.



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3.4.4 Temperature cycling. When tested as specified in 4.4.3.7, the unit shall withstand the temperature cycling without damage and shall meet the performance requirements specified in 3.4.11.

3.4.5 Storage and transit. When tested as specified in 4.4.3.8, the unit shall withstand storage conditions of -80° F. and +155° F. and a simulated altitude of 40,000 feet without damage and shall meet the performance requirements specified in 3.4.11.

3.4.6 Carrying case. The case shall be constructed in clam-shell configuration to provide a rain-resistant closure.

3.4.6.1 Base. The base shall be fabricated from aluminum alloy, as specified in Bill of Material with a wall thickness of .090 inch Brown and Sharp (B & S) gauge No. 11. The base shall be formed with an internal corner radius of 9/16 ± 1/16 inch. The top edge of the base shall be offset inward for strength and to permit the base to be recessed within the cover. The base shall be provided with an accessory clip to secure a 4-ounce oil can; a locking leg brace to maintain a 90° opening of the cover when apparatus is in use; a separate compartment divider for an operating instructions manual; and four rubber bumper-feet located near the corner of the underside of the base. A continuous-type hinge and a drawbolt shall be provided for the proper fastening and closure of the case cover. The base shall be provided with holes to accommodate the motor base plate. The finished base of the assembled unit shall not distort under normal use, or during storage or shipment.

3.4.6.2 Cover. The cover shall be fabricated from aluminum alloy as specified in the Bill of Material with a wall thickness of .090 inch B & S gauge No. 11. The cover shall be formed with an internal corner radius of 9/16 ± 1/16 inch. The lower edge of the cover shall be offset outward for strength and to permit proper mating of the cover with the base. The cover shall be provided with a centrally-located recessed carrying handle, a drawbolt catch, and two bottle holders with integral accessory clip. Provision shall be made for the fastening of the hinge and for locking the leg brace to the cover. The finished cover of the assembled unit shall not distort under normal use, or during storage or shipment.

3.4.7 Handle. The handle shall be smooth, aluminum alloy rod, 1/4 inch diameter, designed to allow a comfortable grip, with a minimum inside hand clearance of 1-1/16 x 3-1/2 inches when in upright position. The handle shall be fastened to the top center with a retaining plate secured by a minimum of four rivets. The longitudinal axis of the handle shall coincide with the longitudinal axis of the case.



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3.4.8 Motor. The pump drive motor shall be the resistance, split-phase type (resistance start-induction run). The motor shall be rated at 1/20-horsepower, 1725 RPM, 115 volts, 60 hertz, 1.30 amps, at continuous time rating, with a maximum temperature rise of 65° C. The motor shall be explosion-proof, class 1, group C, and shall be approved by U.L. The motor shall be given an anti-fungus and tropicalization treatment. The treatment shall consist of a double dip of pigmented epoxy-ester, with glass tie cords on the stator and rust preventative finish on the rotor. No material used in the motor shall support fungus growth.

3.4.9 Motor mount. Provisions shall be made for the mounting of four shock mounts to support the motor assembly on the motor mount and to minimize noise and vibration.

3.4.10 Electric wiring. The unit shall be equipped with a flexible, insulated, three-wire cord, not less than 10 feet in overall length. The cord shall be provided with a grounded cap with standard parallel blades which complies with ANSI C73.11. A permanently marked red warning tag, minimum 0.010 inch thick plastic, shall be attached to the cap end of the cord to indicate that the plug is not explosion-proof. The tag shall contain the following warning: "This plug is not explosion proof and is not intended for use in class I, group C, hazardous locations." An adapter for use with conventional two-wire outlet shall be provided.

3.4.11 Compressor. The compressor shall be a multi-vane rotary pump having a minimum capacity of 0.45 cubic feet per minute. The pump shall be capable of developing a positive pressure of 10 pounds per square inch, and a partial vacuum of 20 inches of mercury. The compressor shall be supplied with an automatic oiler, a suction and pressure filter, a vacuum gauge reading from 0 through 30 inches of mercury, and a vacuum regulator capable of adjustment from 0 through 20 inches vacuum. Performance shall be tested as specified in 4.4.3.5. Unless otherwise specified, visible dome oiler, assembly No. 68 on the drawing, shall be furnished.

3.4.12 Filters.

3.4.12.1 Suction filter. A suction filter shall be provided on the vacuum side of the pump to insure protection of the compressor. The filter shall be freely accessible and shall be visible at all times for inspection. The pressure filter shall be tested in accordance with 4.4.2.1.

3.5 Accessories. All accessories shall be mounted within the case.



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3.5.1 Collection bottles. Collection bottles shall be made of polypropylene, and shall be tested in accordance with 4.4.2.1. The bottles shall have a minimum capacity of 32 fluid ounces.

3.5.1.1 Bottle caps. Each bottle shall be fitted with an airtight rubber closure having metal couplings to fit the corresponding couplings on the drainage tubing. The bottle caps shall be electrically conductive. See Bill of Material for requirements.

3.5.1.2 Bottle holders. Bottle holders, hinge, accessory clips and the drawbolt catch shall be made of corrosion-resisting steel as specified in the Bill of Material.

3.5.2 Drainage tubing. Drainage tubing shall be made of conductive rubber as specified in the Bill of Material and shall be supplied in two, five-foot sections. The tubing shall be 1/4 inch inside diameter with 1/8 ± 1/64 inch wall and shall be provided with couplings fabricated of brass, chromium-plated over nickel. The couplings shall be of the taper-slip type with positive screw locks. The coupling, utilizing a 7/16 inch - 27NS thread, shall insure a tight, leakproof connection. The tubing shall have the following characteristics:

| | |
|---|---|
| Tensile strength (pounds-per-square-inch, minimum) | 1,200 |
| Ultimate elongation (percent, minimum) | 500 |

The tubing shall have a tensile strength and ultimate elongation not less than 75 percent of the original values, after heating by either of the methods specified in 4.4.1.2.1. The electrical resistance of the tubing shall be measured after exposure to oxygen aging, liquid ethyl ether, bending and flexing and oil, as specified in 4.4.1.2.3. Any value in excess of one megohm shall be cause for rejection. Proof of compliance with the electrical resistance requirement shall consist of either labeling by Underwriters' Laboratories, Inc. of the end item, or a laboratory report.

3.5.3 Yankauer cannula. Each unit shall be supplied with one Yankauer cannula (see 6.7). As an alternate, the brass tubing shall conform to alloy 3 of ASTM-B135, the handle material shall conform to composition 260 of QQ-B-613 and the screw machine material shall conform to alloy 360 of QQ-B-626. Manufacturer's name or trademark shall be optional.

3.5.4 Poole abdominal cannula. Each unit shall be supplied with one Poole abdominal cannula (see 6.8). Four to six ferrules on the tubing end of cannula shall be acceptable.



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3.5.5 Spray valve. A spray valve (see 6.6) and tubing shall be supplied with each unit.

3.5.6 Oil. A can of oil, of the same type used in the compressor, with dispenser spout shall be supplied with each unit. When installed, spouts shall be suitably sealed and shall be constructed so as to be opened by perforation, cutting, or other acceptable method. The spout shall be fitted with a small, knurled, threaded or friction cap capable of resealing the end of the spout after opening. Closures, spouts, liners, pads and facings (when used) shall be constructed to provide a leakproof seal. Oil shall be suitable for operation and the lubrication of the compressor at ambient temperatures from 50°F. to 120°F.

3.5.7 Wrench. A combination wrench for use with the suction and pressure filter assembly shall be supplied with each unit.

3.5.8 Shock mounts. Motor shock mounts need not be of conductive rubber if the chassis is properly grounded in accordance with the standards or requirements of the Underwriters' Laboratories, Inc.

3.5.9 Instructions. Printed instructions describing and illustrating expedient means for increasing the drainage capacity and controlling the aspiration rate of the apparatus by the use of additional bottles, tubes, and accessories shall be included. The instructions shall be a part of the service data. The instructions shall include figures 1 and 2, and shall include the following statements.

"Figure 1 illustrates a three suction-bottle series to control waste accumulation.

"Figure 2 illustrates a four suction-bottle series to control aspiration rate and waste accumulation.

"Additional bottles, tops and tubing are not furnished as components of the assembly."

3.6 Weight and dimensions. The weight of the suction and pressure apparatus complete with case and all accessories shall not exceed 33 pounds. The outside dimensions of the complete unit, including top handle, drawbolt, and rubber feet, shall not exceed 14 inches long by 9-1/2 inches wide by 11 inches high.

3.7 Finish.

3.7.1 Accessories. Metal accessories, filters, gauges, control valves, metal couplings, and unpainted interior surfaces shall be chromium-plated over nickel. Corrosion-resistant steel components shall not be plated.



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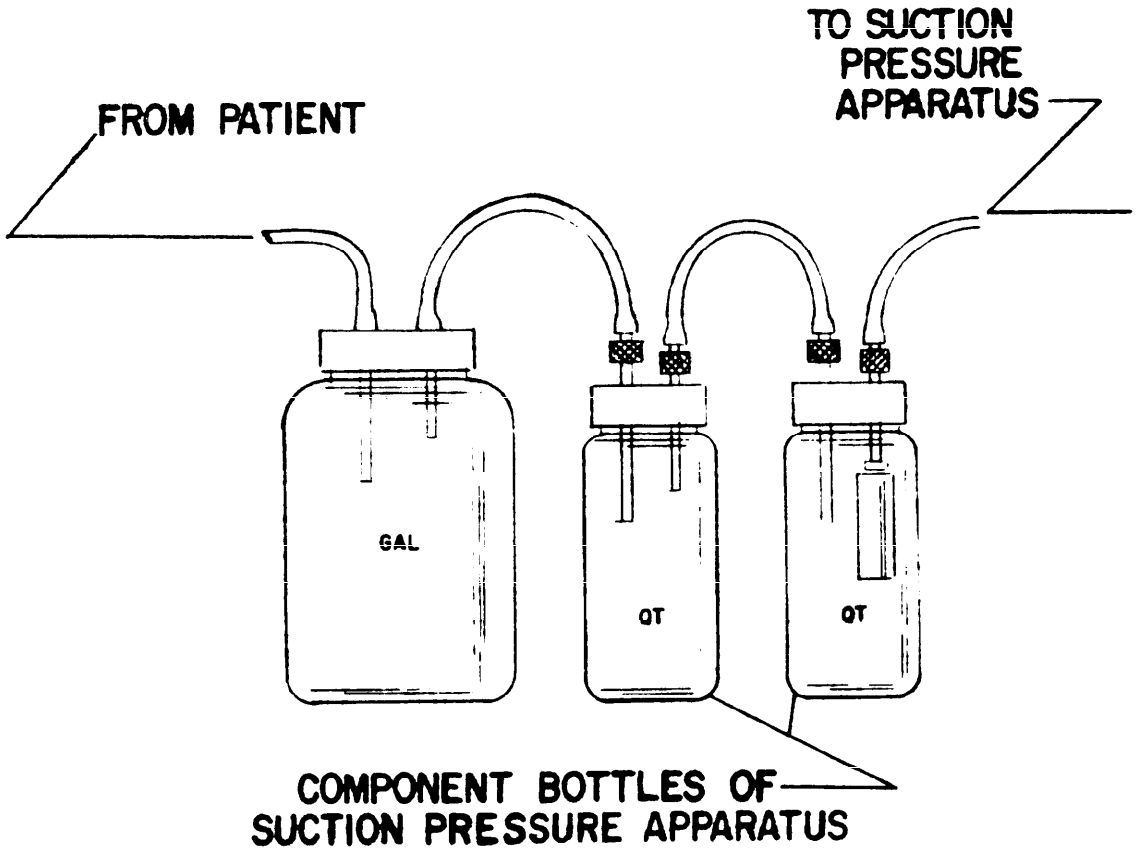



FIGURE 1



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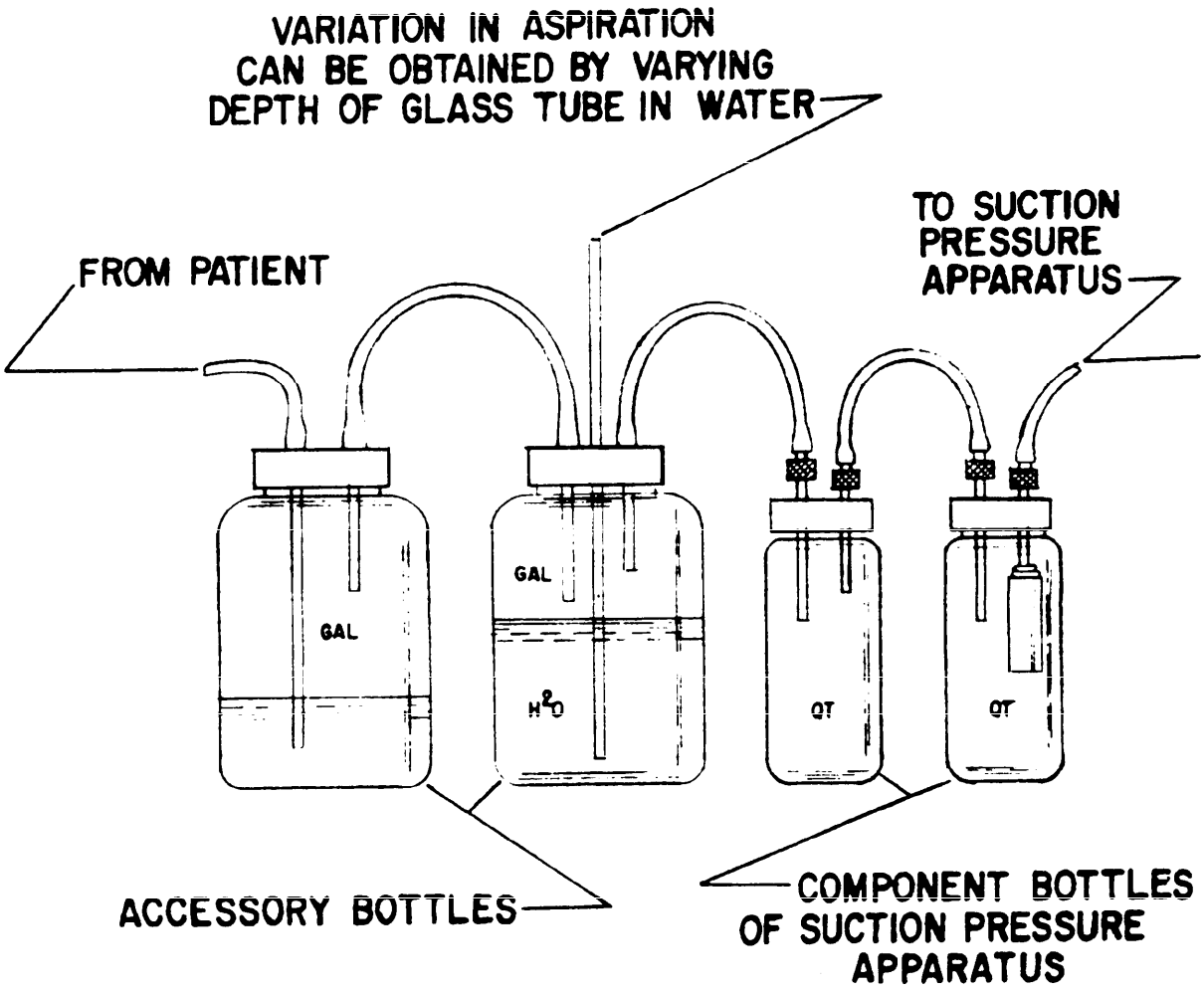



FIGURE 2



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3.7.2 Motor and compressor. Motor and compressor shall be suitably primed and finished with two coats of semi-gloss olive-drab enamel conforming to TT-E-529, and to color 24087 of FED-STD-595.

3.7.3 Base and cover. The base and cover shall be protected against corrosion in accordance with the requirements of MIL-C-5541. All outside and accessible inside surfaces of the base and cover, except the cover support, shall receive a zinc chromate primer conforming to TT-P-1757, which shall be followed by two coats of semi-gloss olive-drab enamel conforming to TT-E-529 and to color 24087 of FED-STD-595. The cover support, part 46 of the Bill of Material, shall be nickel-plated only. All enamel coats shall be sprayed on to accomplish a complete and uniform coverage of the coated parts, and shall be baked to give a smooth, hard and rough finish, which shall be free from runs and resist cracking and chipping.

* 3.8 Identification marking. The carrying case shall be fitted with a metal nameplate, in a conspicuous location. The nameplate shall contain the following data:

National Stock Number
Name or registered trademark of the manufacturer
Serial number
Electrical characteristics
Contract or purchase order number
Oiling instructions

In addition, the motor shall be marked in accordance with MIL-STD-130.

3.9 Service data. The contractor shall furnish, with each unit, two copies of service data containing complete instructions for installation, operation, maintenance, and a list of component parts. The service data shall conform to MIL-M-7298 and shall include such detailed information which shall permit ready installation and maintenance of the item. Service data shall include a cautionary statement that the plug on the line cord is not explosion proof, the plug is not suitable for use in hazardous locations. All information contained in the manual shall be suitably indexed to permit ready reference to the required information. Descriptive material shall include adequate information so that the replacement parts can be ordered by the manufacturer's part number, and without confusion as to the item identification. Identical or interchangeable replacement parts shall be identified by a single part number only. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the contracting officer, two preliminary copies of his proposed service data, for review and comment by Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary service data shall be submitted sufficiently



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early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of required revisions and corrections in the final service data without delaying delivery under the contract or purchase order. The contractor shall mail 15 copies of the approved service data to the contracting officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract, and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon request.

3.10 Workmanship. Workmanship shall be first class throughout. The suction and pressure apparatus shall be free from imperfections which detract from its appearance or impair its serviceability. The finished apparatus shall operate smoothly and noiselessly, and shall perform its functions efficiently, in a manner suitable for the purpose intended.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, and auditory, investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

* 4.1.2.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection.
b. Quality conformance inspection.



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* 4.1.2.1.1 First article inspection. First article inspection shall be performed on one unit when a first article sample is required (see 3.1). This inspection shall include examination of 4.3. and tests of 4.4.

* 4.1.2.1.2 Quality conformance inspection. Quality conformance inspection shall include the examination of 4.3 and tests of 4.4, except 4.4.3.2, 4.4.3.3, 4.4.3.6, 4.4.3.7 and 4.4.3.8.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the component or material may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and Table I.

TABLE I. Sampling for examination

| | Inspection level | AQL (defects per hundred units) |
|---|---|---|
| For visual examination | | |
| Major defects | II | 1.0 |
| Minor defects | II | 4.0 |

4.2.1.1 Dimensional examination. Measurements shall be made on one unit to show conformance to dimensional requirements; measurements of compressor parts shall be made prior to assembly of unit.



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4.2.2 For tests. Sampling for tests shall be selected at random in accordance with MIL-STD-105 (See table II). The acceptance number shall be zero.

TABLE II. Sampling for tests

| Component (unit of product) | Characteristic | Requirement | Test procedure | Inspection level |
|---|---|---|---|---|
| Metals | Chemical composition | Bill of Material | 4.4.1.1 | See 4.4.1.1 |
| Rubber tubing | Physical properties | 3.5.2 | 4.4.1.2.1 | |
| | Electrical conductivity | 3.5.2 | 4.4.1.2.2 | |
| | Electrical resistance | 3.5.2 | 4.4.1.2.3 | |
| Primer | Chemical and physical properties | 3.7 | 4.4.1.3 | See 4.4.1.3 |
| Enamel | Chemical and physical properties | 3.7 | 4.4.1.4 | See 4.4.1.4 |
| Collection bottle | Vacuum | 3.5.1 | 4.4.2.1 | 100 percent |
| Collection bottle cap | Physical and chemical properties | 3.5.1.1 | 4.4.2.3 | S-2 |
| Filter bottle | Vacuum | 3.4.12.1 | 4.4.2.1 | 100 percent |
| Filter bottle | Pressure | 3.4.12.2 | 4.4.2.1 | 100 percent |
| Vacuum gage | Accuracy | Bill of Material | 4.4.2.2 | S-2 |
| End item | Shock and vibration | 3.4.1 | 4.4.3.2 | 1/ |
| | Water resistance | 3.4.2 | 4.4.3.3 | 1/ |
| | Electrical conductivity | 3.4 | 4.4.3.1 | 100 percent |
| | Dielectric strength | 3.4 | 4.4.3.4 | 100 percent |
| | Performance | 3.4.11 | 4.4.3.5 | 100 percent |
| | Humidity | 3.4.3 | 4.4.3.6 | 1/ |
| | Temperature cycling | 3.4.4 | 4.4.3.7 | 1/ |
| | Storage and transit | 3.4.5 | 4.4.3.8 | 1/ |

1/ First article sample only shall be tested. The contractor is to conduct these tests prior to submission to the Government. A test report including a test set up



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description, method of test, data recorded, and conclusion is to accompany the first article. If contractor has previously supplied to the government, an acceptable item with an acceptable test report and proposes to furnish a suction unit identical to that previously furnished, these tests need not be repeated.

4.3 Examination. The suction and pressure apparatus shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with classification of defects listed in table III. Examination shall not be restricted to the classified possible defects listed below.

TABLE III. Classification of defects

| Categories | Defects |
|---|---|
| Major | |
| 101 | Any component part missing. |
| 102 | Exposed edges and corners not free of burs and sharpness. |
| 103 | Any weld omitted; not fused; not free of burns, cracks, and fractures. |
| 104 | Any threaded part stripped, or does not function smoothly. |
| 105 | Cover warped, or otherwise deformed; does not close flush. |
| 106 | Unit not clean; not free of dirt, oil, and grease. |
| 107 | Sound-deadening material missing. |
| 108 | Any bottle not free of cracks, breaks, chips, and sharp and deformed areas. |
| 109 | Tapered slip-type couplings do not mate properly. |
| 110 | Oil leak from pump, or oil can. |
| 111 | Warning tag on electrical cord not as specified. |
| Minor | |
| 201 | Blistering, checking or peeling of the finish. |
| 202 | Any screw not drawn up tight; head not flush with the surface. |
| 203 | Color of enamel finish not as specified. |
| 204 | Service data not included. |
| 205 | Identification marking missing, incomplete, illegible, or not properly attached. |
| 206 | Finish not free of tackiness, soft runs, and sags. |
| 207 | Finish not free of scratches, bare spots, and areas of visibly thin coating. |
| 208 | Rivets not properly installed. |



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4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Materials.

* 4.4.1.1 Chemical composition. Tests for chemical analysis shall be conducted in accordance with method 111 or 112 of FED-TEST METHOD STD-151. Four ounces from each lot of metal used shall be selected for test.

* 4.4.1.2 Rubber tubing. Two samples of rubber tubing representative of each lot used in the manufacture of the apparatus shall be selected for test. Each sample shall be large enough to provide the required number of specimens for the tensile tests, both before and after aging, and for the electrical resistance tests.

4.4.1.2.1 Physical properties. Physical properties shall be determined in accordance with the appropriate test method of FED-TEST METHOD STD-601 in table IV.

TABLE IV. Physical properties

| Property | Test method |
|---|---|
| Tensile strength | 4111 |
| Ultimate elongation | 4121 |
| Oxygen pressure-heat | 7111 |
| Air heat | 7221 |

4.4.1.2.2 Electrical conductivity. The rubber tubing shall be tested for electrical conductivity in accordance with the applicable procedures outlined in NFPA No. 56A.

4.4.1.2.3 Electrical conductivity after tests. The electrical conductivity of the rubber tubing shall be determined after being subjected to the following tests.

4.4.1.2.3.1 Oxygen aging. Two 7-inch lengths of tubing shall be subjected to oxygen under pressure of 300 psi for a period of 94 hours at a temperature of 70° C. as specified in 7111 of FED-TEST METHOD STD-601. The electrical resistance of the tubing shall be measured before and after the oxygen aging.



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4.4.1.2.3.2 Liquid ethyl ether. Two 5-foot lengths of tubing shall be maintained for 5 hours full of liquid ethyl ether. The electrical resistance of the tubing shall be measured initially and at 1/2 hour intervals during the tests.

4.4.1.2.3.3 Bending and flexing. Two 5-foot lengths of tubing, fitted with electrical test connections, shall be passed back and forth repeatedly on a pulley 1-3/4 inches in diameter as specified in U.L. 1067. The tubing shall be held in tension by a 5-pound weight suspended from the free end of the sample. The electrical resistance of the tubing shall be measured initially and at intervals during the test.

4.4.1.2.3.4 Oil. Two 7-inch lengths of tubing shall be submerged for 5 days in SAE 30 motor oil. The resistance of the tubing shall be measured before and after the test.

4.4.1.3 Primer. Primer shall be tested in accordance with the quality assurance provisions of TT-P-1757 and MIL-C-5541.

4.4.1.4 Enamel. Tests on enamel shall be conducted in accordance with the procedures outlined in TT-E-529.

4.4.2 Components.

4.4.2.1 Bottles. The collection bottle and the suction filter bottle shall be subjected to an internal vacuum of 25 inches of mercury for a period of one minute without failure. The pressure filter bottle or sight glass shall be subjected to an internal pressure of 30 psig for a period of one minute without failure. Vacuum shall be applied during test specified in 4.4.3.5.

* 4.4.2.2 Gauges. The gauges shall be measured for accuracy using standard gauges accurate to the nearest 1/8 inch of mercury. Not less than 3 samples shall be tested regardless of lot size.

4.4.2.3 Collection bottle cap.

4.4.2.3.1 Electrical conductivity. Test for electrical conductivity shall be as outlined in NFPA No. 56A.

4.4.2.3.2 Hardness. Hardness shall be conducted as outlined in FED-TEST METHOD STD-601, Method No. 3021, using a shore A durometer.

4.4.3 End item.



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4.4.3.1 Electrical conductivity. The apparatus, complete with rubber feet and rubber tubing, shall be tested in accordance with the applicable procedures outlined in NFPA No. 56A.

4.4.3.2 Shock and vibration. The unit, with all components inserted in place within the case, shall be dropped from a height of 36 inches onto dry earth. Repeat this drop test so that the unit is dropped on each surface of the case. After the unit has been dropped on each surface of the case, subject the unit to a vibration load at a fixed frequency between 3 and 5 cycles per second with an amplitude of displacement of approximately 1-3/4 inches in both the horizontal and vertical directions for a five minute period. If no damage is observed after vibration testing, the unit shall be operated for 4 hours and shall then be tested in accordance with 4.4.3.5.

4.4.3.3 Water resistance. The unit, in its case, shall be subjected to a shower at the rate of 3 inches per hour for a period of 5 minutes. The contents shall be examined to determine if moisture has entered the case.

* 4.4.3.4 Dielectric strength. A dielectric test potential of 1000 volts, plus twice the rated voltage, 60 Hertz, single-phase, alternating current, shall be applied for 1 minute between the exposed noncurrent-carrying parts and the current-carrying parts in the circuit. Test shall be performed after the motor has been operating for 10 - 15 minutes and while it is still warm.

4.4.3.5 Performance. The unit shall be operated in accordance with the manufacturer's instructions. The following determinations shall be made after operating at least 10 minutes:

Delivery at pressure end (cubic feet per minute and psi).
Intake at suction end (inches of mercury).

4.4.3.6 Humidity. At a relative humidity of 95 ± 5 percent, cycle the unit as follows: Raise temperature linearly from 80°F. to 115°F. in 6 hours; maintain at 115°F. for 6 hours; decrease temperature linearly from 115°F. to 80°F. in 6 hours, maintain at 80°F. for 6 hours. Then test the unit for performance as specified in 4.4.3.5 after a 5 minute and a 4 hour operating time.

4.4.3.7 Temperature cycling. After conditioning the unit at 120°F. for 8 hours, operate the unit at 120°F. for 4 hours. Then, after conditioning the unit at -60°F. for 8 hours, allow the temperature to rise to 50°F. and operate for 4 hours at 50°F. The unit shall be tested for performance as specified in 4.4.3.5 after a five minute and a four hour operating period at both the 120°F. and 50°F. conditioning temperatures.



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4.4.3.8 Storage and transit.

a. After storing the unit at -80°F. for three days, raise the temperature to 70°F. and operate for 4 hours.
b. Store the unit at + 155°F. for three days, lower the temperature to 70°F. and again operate for 4 hours.
c. Subject the unit to a simulated 40,000 feet altitude (5.0 inches of mercury) for 5 hours, return to normal atmospheric pressure and operate for four hours.
d. Test the unit for performance as specified in 4.4.3.5 after a five minute and a four hour operating period for each test condition a, b, and c.

4.5 Packaging inspection. The inspection of the packaging and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING.

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each suction and pressure apparatus along with accessories shall be positioned within their respective compartments in the carrying case. The case shall be closed and latched. The latched case shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-601, overseas type or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

* 5.1.1.1 Case liner. Wood boxes shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed in accordance with the appendix of PPP-B-636.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.1.2 Level C. Each unit shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, constructed so as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.



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* 5.2 Marking. Exterior containers shall be marked in accordance with MIL-STD-129. MIL-STD-129 markings shall not be placed on the case. The level of protection shall be shown as "A" in lieu of "A/A" or "C" in lieu of "C/C" on the exterior container.

6. NOTES

* 6.1 Intended use. This unit provides suction for patient aspiration, removal of fluids during surgery, and testing of implanted drainage tubes; a pressure valve is provided for spray applications. Because of its small size and self-contained, field type carrying case, it is particularly useful in field medical activities. The unit is provided with a non-explosion proof, electrical plug which can only be used outside the hazardous area or above the five-foot level in the hazardous area. An explosion-proof electrical plug must be attached for direct use in a hazardous area containing flammable gases.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

a. Title, number and date of this specification.
b. Selection of applicable levels of packaging (see 5.1).
c. When a first article is required (see 3.1, 4.1.2.1 and 6.3).

* 6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of the Armed Services Procurement Regulations. The contract or purchase order specifies the quantity, procedure for delivery and other specific instructions for inspection and approval of the first article.

6.4 This specification covers the following item appearing in the Federal Supply Catalog:

| National Stock Number | Item Identification |
|---|---|
| 6515-00-782-2625 | Suction and Pressure Apparatus, Surgical, Explosion-Proof, Portable, Field, 115 volt, 60 Hz, A.C. |

6.5 An illustration of the Suction and Pressure Apparatus appears as figure 3.

6.6 The spray valve of the type required in paragraph 3.5.5 is available from Phillips Drucker Co. P/N M110B, J. Sklar Mfg. Co. P/N 100-630, or Vernitron Medical Products - P/N 2112-2.

6.7 Yankauer cannula shall conform to requirements for National Stock Number 6515-00-386-7600.



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. 6.8 Poole abdominal cannula shall conform to requirements for National Stock Number 6515-00-386-6600.

6.9 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of the document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

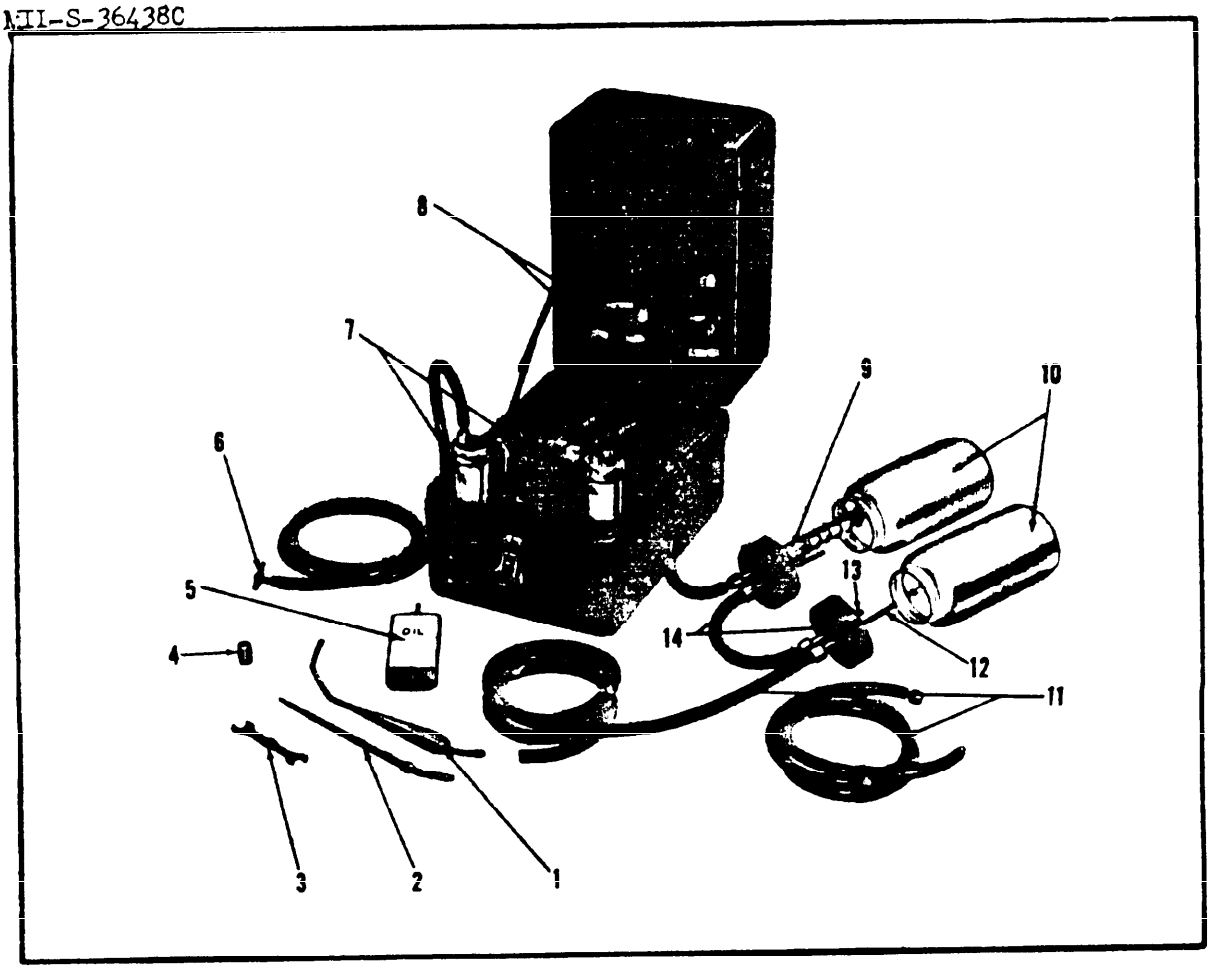
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1. CANNULA, LARYNGEAL, Yankauer
2. CANNULA, ABDOMINAL, Poole
3. WRENCH, OPEN END
4. MUFFLER, EXHAUST
5. LUBRICATING OIL
6. SPRAY VALVE
7. JAR, POLYPROPYLENE
8. DISK, RUBBER
9. OVERFLOW CUTOFF ASSEMBLY
10. BOTTLE, POLYPROPYLENE
11. TUBING, RUBBER, Conductive
12. TUBE ASSEMBLY, METAL, Straight, Female
13. TUBE ASSEMBLY, METAL, Straight, Male
14. CAP, Conductive Rubber

FIGURE 3. Suction and pressure apparatus



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OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-S-36438C - SUCTION AND PRESSURE APPARATUS, SURGICAL, Explosion-Proof, Portable, Field

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

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